

WILLISTON SLOULIN FIELD, ND, USA

WMO: 727670

Lat: 48.174N

Lon: 103.637W

Elev: 1902

StdP: 13.71

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94014

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.4	-16.0	-27.2	1.3	-20.5	-21.9	1.8	-15.4	31.0	18.2	26.8	21.2	4.9	200	0.667

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.4	93.7	67.4	89.9	66.5	86.4	65.4	71.4	85.6	69.4	83.8	67.7	81.4	10.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.0	106.6	77.8	64.6	98.0	76.3	62.6	90.9	74.3	36.4	85.9	34.7	83.6	33.0	81.4	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.7	23.6	20.2	DB	-27.5	100.6	5.6	2.9	-31.5	102.7	-34.8	104.4	-38.0	106.0	-42.1	108.1	(4)
(5)			WB	-27.5	75.5	5.4	2.5	-31.4	77.3	-34.5	78.8	-37.6	80.2	-41.5	82.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	42.1	11.9	15.5	28.4	42.7	53.7	63.6	71.0	69.3	58.6	44.0	28.3	16.2	(6)	
(7)		DBStd	23.33	15.25	14.29	14.04	10.00	8.10	7.00	6.08	6.65	8.60	9.40	11.99	13.65	(7)	
(8)		HDD50	5081	1181	965	671	256	54	3	0	0	22	230	650	1049	(8)	
(9)		HDD65	8839	1646	1385	1133	670	361	104	15	32	225	654	1100	1514	(9)	
(10)		CDD50	2193	0	0	3	36	167	411	652	599	281	43	1	0	(10)	
(11)		CDD65	475	0	0	0	0	9	63	202	166	34	1	0	0	(11)	
(12)		CDH74	6505	0	0	1	38	221	890	2499	2178	633	45	0	0	(12)	
(13)		CDH80	2664	0	0	0	7	53	286	1095	973	242	8	0	0	(13)	
(14)	Wind	WSAvg	8.7	8.3	8.4	9.1	10.0	10.0	9.1	8.2	8.0	8.2	8.7	8.2	8.0	(14)	
(15)	Precipitation	PrecAvg	14.4	0.6	0.4	0.7	1.2	2.1	2.5	2.2	1.4	1.2	0.9	0.6	0.6	(15)	
(16)		PrecMax	21.9	1.8	1.7	2.2	3.3	7.4	6.2	6.6	4.6	3.5	3.5	3.3	2.0	(16)	
(17)		PrecMin	9.1	0.0	0.0	0.0	0.0	0.2	0.4	0.2	0.0	0.0	0.0	0.0	0.0	(17)	
(18)		PrecStd	3.3	0.4	0.4	0.5	0.8	1.4	1.3	1.5	1.0	0.9	0.7	0.5	0.4	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.3	48.7	68.2	79.5	86.1	92.9	98.9	98.8	93.3	79.8	62.8	47.2	(19)	
(20)			MCWB	38.8	39.9	50.2	55.9	61.4	68.2	68.9	67.5	65.3	57.3	48.7	38.9	(20)	
(21)		2%	DB	39.5	42.4	59.3	71.3	80.0	87.1	93.9	93.7	86.8	71.6	55.1	41.2	(21)	
(22)			MCWB	34.7	36.6	45.5	52.2	59.3	65.8	68.3	66.8	62.8	54.4	43.9	35.6	(22)	
(23)		5%	DB	36.5	38.3	52.6	66.0	75.3	82.8	90.0	89.6	81.4	65.2	49.6	37.5	(23)	
(24)			MCWB	33.0	34.2	42.4	49.5	57.0	64.1	67.7	65.6	61.0	50.7	40.7	33.2	(24)	
(25)		10%	DB	33.1	34.5	46.6	61.1	70.8	79.1	86.3	85.4	76.0	59.6	44.2	34.2	(25)	
(26)			MCWB	30.5	31.8	39.3	47.0	54.9	62.2	66.9	64.7	58.9	48.4	37.7	30.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.7	40.8	50.6	56.7	64.5	72.9	75.4	73.0	68.1	59.8	49.5	39.4	(27)	
(28)			MCD B	44.5	47.2	66.4	74.9	78.2	86.4	88.1	87.6	84.3	75.3	61.6	45.7	(28)	
(29)		2%	WB	35.2	37.0	46.3	53.3	61.4	69.1	72.4	70.1	65.3	55.5	44.7	36.1	(29)	
(30)			MCD B	38.9	42.0	58.0	68.6	75.6	81.5	86.5	86.5	81.3	68.7	54.1	40.7	(30)	
(31)		5%	WB	33.1	34.4	42.8	50.5	58.8	66.3	70.3	68.3	62.7	52.1	41.2	33.5	(31)	
(32)			MCD B	36.4	37.8	51.9	64.2	71.5	78.8	84.1	83.6	77.5	62.9	48.8	37.2	(32)	
(33)		10%	WB	30.5	32.0	39.6	47.9	56.5	64.1	68.5	66.4	60.3	49.2	38.1	31.0	(33)	
(34)			MCD B	33.2	34.8	46.1	59.8	68.3	76.3	82.0	81.1	73.6	58.8	43.8	34.1	(34)	
(35)	Mean Daily Temperature Range	MDBR		19.1	19.3	20.7	24.1	24.9	24.2	26.4	27.9	26.9	23.0	20.0	18.3	(35)	
(36)		5% DB	MCDBR		20.3	21.7	30.2	33.5	32.9	30.7	31.7	33.8	36.9	33.9	28.4	21.4	(36)
(37)			MCWBR		16.5	16.4	18.2	17.9	16.1	14.0	12.3	13.0	16.5	18.4	16.5	(37)	
(38)		5% WB	MCDBR		19.8	20.5	28.8	31.2	27.9	26.3	27.0	28.9	31.4	31.3	27.2	21.1	(38)
(39)			MCWBR		16.4	15.7	17.9	17.2	14.6	13.6	12.0	12.4	14.7	17.2	18.0	16.6	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.253	0.264	0.278	0.326	0.341	0.356	0.365	0.374	0.328	0.294	0.262	0.248	(40)	
(41)		taud		2.445	2.462	2.527	2.405	2.423	2.423	2.419	2.368	2.465	2.549	2.519	2.473	(41)	
(42)		Ebn at Noon		268	290	303	294	292	287	283	275	279	275	264	257	(42)	
(43)		Edh at Noon		20	24	27	34	35	35	35	35	29	23	18	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg		486	812	1211	1547	1845	2021	2157	1763	1305	810	493	352	(44)	
(45)		RadStd		46	84	99	89	141	144	96	115	113	86	64	25	(45)	

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days				
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page